

Combinatorial Theory: Positroids / 9.21.26

Last week, we discussed



Assume our field $\mathbb{F} = \mathbb{R}$. Then we have

→ Key Fact: the coordinate ring of $Gr(2, n)$
 = Ptolemy's Algebra

$$P \cong P_n$$

$$= \mathbb{R}[\Delta_{ij}]$$

$$\text{for } 1 \leq i < j \leq n$$

(for $i < j < k < l$)

three term
Plücker
relation!

$$\left\langle \Delta_{ik} \Delta_{jl} = \Delta_{ij} \Delta_{kl} + \Delta_{il} \Delta_{jk} \right\rangle$$

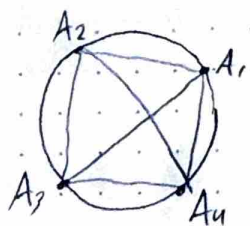
Recall that for a matrix A and determinant Δ_{ij}

$$A = \begin{bmatrix} a_1 & \dots & a_n \\ b_1 & \dots & b_n \end{bmatrix}$$

$$\Delta_{ij} = \begin{vmatrix} a_i & \dots & a_n \\ b_i & \dots & b_n \end{vmatrix}$$

Ptolemy's Thm:

$$|A_1 - A_3| \cdot |A_2 - A_4| = |A_1 - A_2| \cdot |A_3 - A_4| + |A_1 - A_4| \cdot |A_2 - A_3|$$

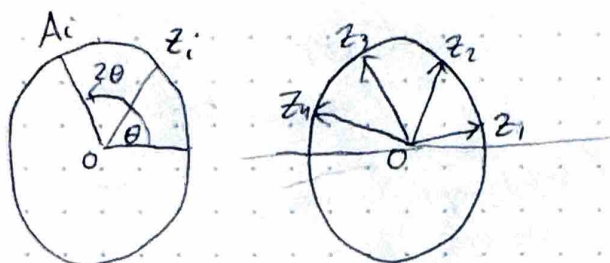


$A_1 A_2 A_3 A_4 \in \text{Unit cycle in } \mathbb{C}$

we can now give a proof of Ptolemy's Thm - (see relationship to three-term Plücker rel.)

→ pick $z_1, z_2, z_3, z_4 \in \mathbb{C}$, such that

$$(z_i)^2 = A_i, \quad |z_i| = 1, \quad \text{Im}(z_i) \geq 0.$$



Let $z_i = a_i + \sqrt{-1} b_i$. Let's do the following manipulations

$$A_i = \frac{(z_i)^2}{|z_i|^2} = \frac{(z_i)^2}{z_i \cdot \bar{z}_i} = \frac{z_i}{\bar{z}_i}$$

$$|A_i - A_j| = \left| \frac{z_i}{\bar{z}_i} - \frac{z_j}{\bar{z}_j} \right|$$

purely imaginary

$$= \left| \frac{z_i \bar{z}_j - z_j \bar{z}_i}{\bar{z}_i \bar{z}_j} \right| = |z_i \bar{z}_j - z_j \bar{z}_i|$$

$$= 2 |\det \begin{pmatrix} a_i & a_j \\ b_i & b_j \end{pmatrix}|$$

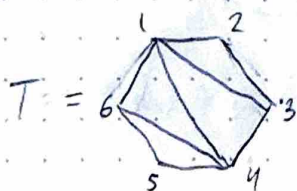
$$= 2 |\Delta_{ij}|.$$

we still need to ensure $\Delta_{ij} > 0$, which will be true if $i < j$.

↳ Plücker-relations $\Rightarrow$ Ptolemy.
(3-term)

(Recall) let T be a triangulation of an n -gon. Then a "cluster" contains Plücker coordinates that describe a triangulation.

EX:



$$T =$$

$$C_T = \left\{ \overbrace{\Delta_{12}, \Delta_{23}, \dots, \Delta_{n-1n}, \Delta_{n1}}^{\text{boundary of } \square}, \underbrace{\Delta_{15}, \Delta_{14}, \Delta_{45}}_{\text{inner-edges of triangulation } T} \right\}$$

Generally, we can describe a cluster C_T for triangulation T as

$$C_T = \left\{ \Delta_{ij} \mid (i,j) \text{ is an edge in } T \right\}$$

Thm:

→ Fix a triangulation T . Then we have

- 1) any Δ_{ij} can be expressed in terms of variables in C_T by a subtraction-free rational expression
- 2) variables in C_T are algebraically independent
- 3) for any positive real valued _____ of Plücker coords in C_T , there exists a unique cell (Δ_{ij}) satisfying Plücker relations.

(Recall.) for positive-real-valued frieze patterns of order n

$$P = (p_{ij}), \quad i, j \in \mathbb{Z}, \quad i < j < i+n$$

we have the following

- 1) $p_{i,i+1} = p_{i+n,i+1} = 1$
- 2) $p_{ij} \in \mathbb{R}_{>0}$
- 3) diamond relations

$$\begin{vmatrix} p_{ij} & p_{i,j+1} \\ p_{i+1,j} & p_{i+1,j+1} \end{vmatrix} = 1$$

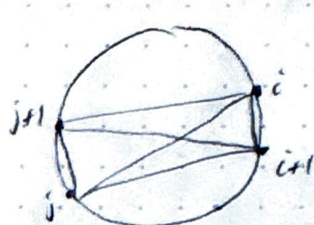
Plücker relations
 $\Delta_{12} \Delta_{23} = \dots \Delta_{n-1,n} \Delta_n = 1$

Thm:

$$\left\{ \begin{array}{l} \text{frieze patterns of} \\ \text{order } n \end{array} \right\} \xleftrightarrow{\text{bijection}} \left\{ \frac{(\Delta_{ij})}{\Delta_{ij} > 0} \right\}$$

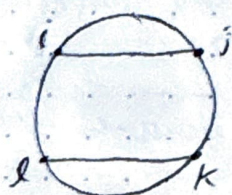
namely, $p_{ij} \mapsto \Delta_{ij}$ for $1 \leq i < j \leq n$

the point here is that "diamond relations" $\Leftrightarrow$ special 3-term Plücker relations
 $\hookrightarrow$ not all 3-term Plücker relations!



$$\Delta_{ij} \cdot \Delta_{i+1, j+1} = \Delta_{i, j+1} \cdot \Delta_{i+1, j} + \Delta_{i, i+1} \cdot \Delta_{j, j+1}$$

here's one where this fails:



$$\Delta_{ij} = \Delta_{kl} = 1, \text{ all others are zero}$$

$\hookrightarrow$ Plücker rel. fails, as $0 \neq 1$.

$\rightarrow$ all "special" 3-term relations hold

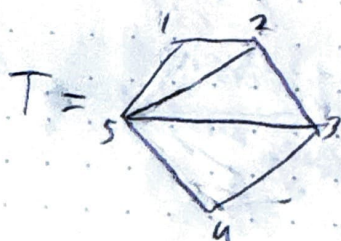
Lemma (exercise):

$\hookrightarrow$ if we assume that all $\Delta_{i,j} > 0$, then

special 3-term rels. $\Rightarrow$ all 3-term rels.

Lets now return to constructing frieze patterns

$\hookrightarrow$ Consider a triangulation T , and the associated cluster $\mathcal{C}_T$ of Plücker relations.

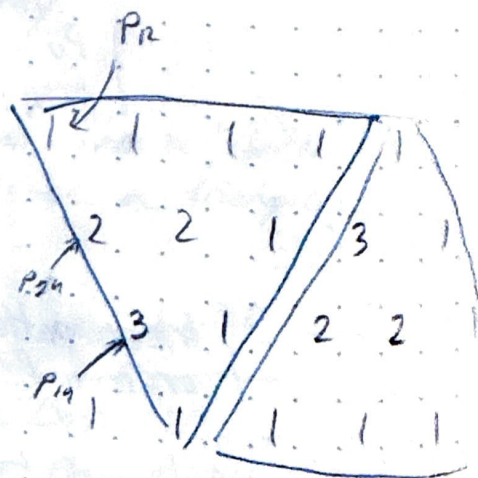


$$\Delta_{12} = \dots = 1$$

$$\Delta_{13} = 2$$

$$\Delta_{24} = 2$$

$$\Delta_{14} = 3$$



We can also consider a construction by Conway and Coxeter

↳ for $i=1, 2, \dots, n$ vertices in your triangulation

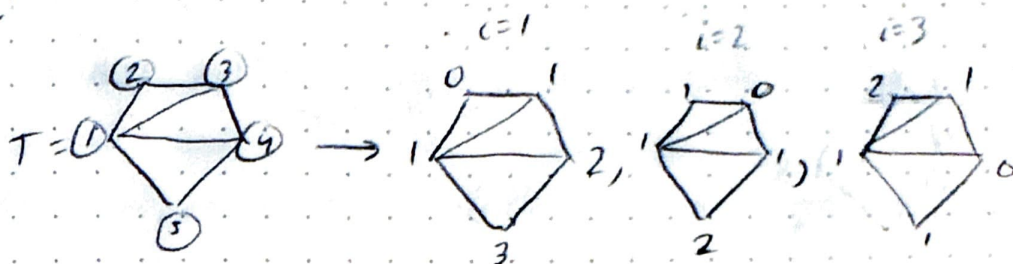
→ label vertex i by 0

→ label all neighbors of i by 1

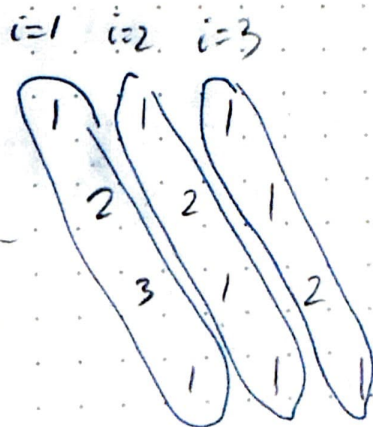
→ for all triangles in T ,
if two vertices are labeled
by a and b , label the
third as $a+b$.



Ex:



then we construct

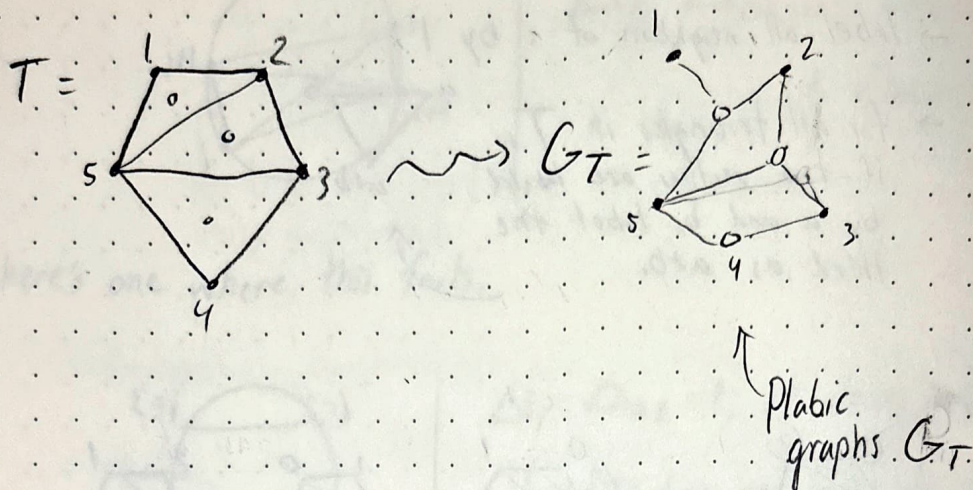


Both methods recover the same pattern, given a fixed T .

↑ (exercise: show this)

Both previous construction methods were recursive.

↳ what else could we do



→ # of "almost perfect" (has $n-2$ edges.) of G_T that include all vertices except i, j , will correspond to Plücker coordinate P_{ij} for example:

$$P_{12} = 1$$

$$P_{14} = 2$$

$$P_{13} = 3$$

